



On Enhancing the Performance of Children's ASR Task in Limited Data Scenario

Ankita ¹, Shambhavi ², Syed Shahnawazuddin¹

¹Department of ECE, NIT Patna, India

²Computer Science Department, IIT Patna, India

ankita.ph21.ec@nitp.ac.in, shambhavi_2311ai39@iitp.ac.in, s.syed@nitp.ac.in

Abstract

In this paper, we have documented our efforts towards developing a robust children's automatic speech recognition (ASR) system in limited data scenario. At first, we have explored the effect of in-domain data augmentation so as to deal with limitations posed by data scarcity. This also helps in developing a competitive baseline ASR system. Next, we have studied the affect of modeling glottal activity parameters along with spectrum-based front-end acoustic features like the Mel-frequency cepstral coefficients (MFCC). Finally, the impact of feature normalization through feature-space maximum likelihood linear regression (fMLLR) is explored. As a consequence of applying fMLLR and then concatenating the normalized MFCC features with glottal activity parameters, a relative reduction in character error rate by 40% over the baseline is obtained.

Index Terms: Children's ASR, glottal activity parameter, feature normalization, fMLLR.

1. Introduction

In this paper, we present our efforts towards developing an effective automatic speech recognition (ASR) system for children in limited-data conditions using end-to-end (E2E) architecture. It is well known that, E2E architectures are extremely data hungry [1–6]. Hence, scarcity of domain-specific data is the first major hiccup in the development of an efficient ASR system for children. This fact has also been highlighted in some of the earlier works on children's ASR [7–10]. Employing data augmentation helps overcome this issue. Therefore, in this study, we have first resorted to in-domain data augmentation to deal with the ill-effects of data scarcity. As a consequence, the developed children's ASR system yielded error-rates that were competitive to those reported in recent works [8–12].

In order to further boost the recognition rates, we have first studied and documented the impact of feature normalization. We have employed feature-space maximum likelihood linear regression (fMLLR) for this purpose. The employed fMLLR transforms are optimized through speaker adaptive training (SAT). Hidden Markov models (HMM), with observation densities generated through Gaussian mixture models (GMM), was used for optimizing the fMLLR transforms. Feature normalization through fMLLR helps in reducing the ill-effects inter-speaker variability. Next, we have studied the impact of modeling the glottal activity (GA) parameters as done in [10]. The explored GA parameters are strength of excitation (SoE), normalized auto-correlation peak strength (NAPS), and higher-order statistics (HOS). The GA parameters are fused with the Mel-frequency cepstral coefficients (MFCC) at the short-time frame level. As argued and demonstrated in [10], feature con-

catenation helps in capturing a wider range of relevant acoustic attributes. This, in turn, leads to reduction in error rates. Finally, in our proposed approach, the SoE parameter is fused with the fMLLR-normalized MFCC features. In other words, the proposed idea involves appending the fMLLR-normalized MFCC features and SoE parameter at the short-time frame level. The resultant feature vectors are then used for the training purpose. Since, the explored glottal activity parameters exhibit very low canonical correlation when compared with the MFCC features, it can be concluded that these parameters capture acoustic information missed out by MFCC. Furthermore, an analysis presented later in this paper reveals that the explored GA parameters show very little variation with age and gender of the child speakers. Motivated by these facts, we concatenated the fMLLR-normalized MFCC features and GA parameters in order to capture a wider range of relevant acoustic information as well as to enhance robustness towards age and gender variations. Subsequently, significant reductions in error-rates over the baseline were noted.

It is worth highlighting here that, even though fMLLR-based normalization is a very commonly used technique, its impact on children's ASR employing E2E architecture has not been studied yet. In [13], the role of vocal tract-length normalization (VTLN) was employed for improving children's speech recognition. However, in our experimental exploration, VTLN was not observed to be very effective. This may be due to the fact that no children's data was used for training in [13]. The acoustic mismatch between the training and testing data was reduced by the use of VTLN. Similarly, the use of GA parameters has been studied and documented in [10]. However, the effect of concatenating SoE parameter with fMLLR-normalized MFCC features has not been done before.

The rest of this paper is organised as follows: In Section 2, the proposed idea is discussed in detail. The experimental evaluations demonstrating the effectiveness of the proposed approach are presented in Section 3. Finally, the paper is concluded in Section 4.

2. Proposed idea

The proposed idea to enhance children's speech recognition is pictorially depicted in Fig. 1. It consists of three phases, namely, in-domain data augmentation, fMLLR-based feature normalization and feature concatenation. Each of these phases are described in detail in the following subsections.

2.1. In-domain data augmentation

As discussed earlier, scarcity of domain-specific data is the first major hiccup while developing an ASR system for children. It is worth highlighting here that, unlike adults' speech case, chil-

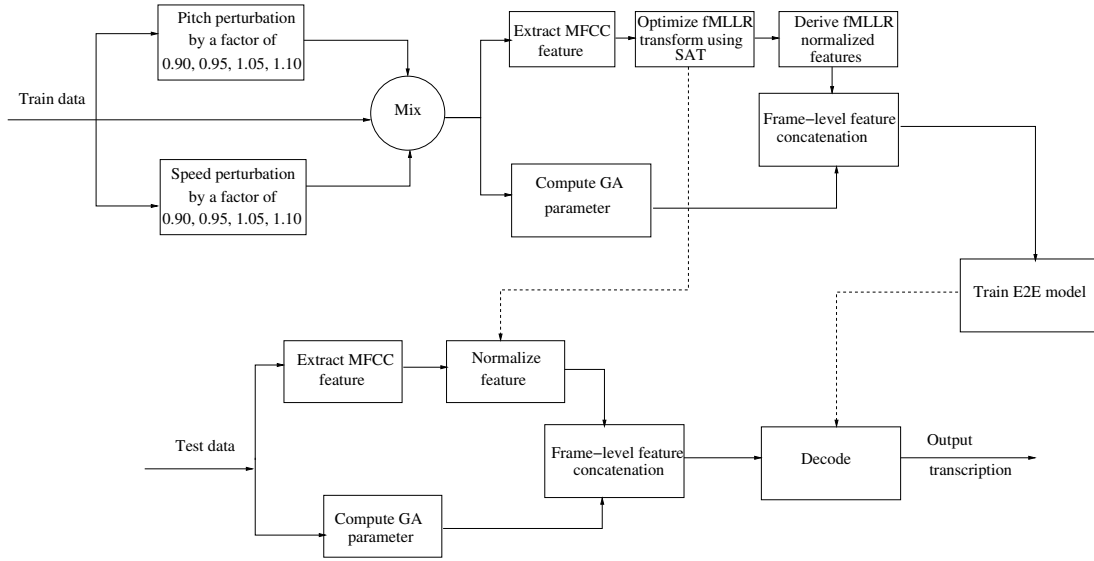


Figure 1: Block diagram summarizing the proposed idea.

dren’s speech data is available in a handful of languages. At the same time, the amount of available data is just a few tens of hours. Paucity of data leads to significantly lower recognition rates in the case of children’s ASR especially when E2E architectures are employed. Therefore, we have resorted to data augmentation to overcome the issues related to data scarcity.

The primary motive of employing in-domain data augmentation is to increase the amount of children’s speech for training. For that purpose, speed and pitch perturbation are performed to create newer versions with slight change in duration and pitch when compared to the original training data-set. The scaling factors are chosen as 0.90, 0.95, 1.05, 1.10 in both the cases. As a consequence, 8 newer versions of the available children’s training speech data are created. These newer versions are pooled into training along with the unperturbed data. This leads to a substantial increase in the amount of training data. At the same time, we are able to capture greater diversity by applying speed and pitch perturbation. As already mentioned earlier, a competitive baseline recognition performance was obtained as a consequence of in-domain data augmentation.

2.2. Explored glottal activity parameters

At the time of voiced speech production, quick movement of vocal folds results in high energy at the closing phase of glottis. It imparts elevated strength to the voiced speech at the locations of epochs. The strength estimation at the epochs are done by passing the speech signal through zero-frequency filter (ZFF). Next, the slope of the filtered signal is calculated near the glottal closure instants (GCI) or epochs [14]. The absolute value of slope yields the desired strength of excitation $s_e[n]$. SoE is given by:

$$s_e[n] = |z[n+1] - z[n]| \quad (1)$$

where, $z[n]$ is the zero-frequency filtered signal at the n th epoch location.

For the production of voiced sounds, the time-varying vocal-tract system is excited by quasi-periodic glottal pulses of air from the lungs. The speech signal thus produced is therefore quasi-periodic for voiced sounds and this behaviour can be characterised by taking the normalized autocorrelation peak

strength of signal $z[n]$. The NAPS $n_p(\sigma)$ for a frame of filtered signal $z[n]$ is given by:

$$n_p(\sigma) = \frac{\sum_{n=1}^N z[n] * z[n - \sigma]}{\sum_{n=1}^N z^2[n]} \quad (2)$$

where, σ is the time-lag which represents the location of the highest peak in 1.5-2.5 ms range.

While producing voiced speech, the glottal opening and closing instances results in significant excitation due to the air-flow pressure. There is a higher strength during the closing instances compared to that of opening due to differences in the airflow pressure around those instants. Consequently, the shape of the glottal pulse turns out to be asymmetric [15]. The power of skewness-to-kurtosis ratio (skr) derived from the integrated linear prediction residual (ILPR) signal is used for capturing the asymmetric characteristics of the glottal pulse and is given by:

$$skr = \frac{\alpha^2}{\beta^{1.5}} \quad (3)$$

α is the skewness of the source signal given as:

$$\alpha = \frac{\frac{1}{N} \sum_{n=1}^N (r_i[n] - \bar{r}_i)^3}{\left(\frac{1}{N} \sum_{n=1}^N (r_i[n] - \bar{r}_i)^2 \right)^{\frac{3}{2}}} \quad (4)$$

where, $\bar{r}_i$ is the mean of $r_i[n]$ (the ILPR signal). On the other hand, β is the kurtosis calculated as:

$$\beta = \frac{\frac{1}{N} \sum_{n=1}^N (r_i[n] - \bar{r}_i)^4}{\left(\frac{1}{N} \sum_{n=1}^N (r_i[n] - \bar{r}_i)^2 \right)^2} - 3 \quad (5)$$

2.3. Motivation for using glottal activity parameters

As mentioned earlier, one of the contributions of this work is to study the effect of glottal activity parameters on low-resource children’s ASR task. In this regard, the first thing to do was to determine whether these parameters are suitable for children’s

Table 1: Mean and variance values for the different glottal activity parameters studied in this paper. The values are tabulated with respect to the age-group and gender of the child speakers.

Glottal activity parameter	6-7 Years		8-9 Years		Female		Male	
	Mean	Variance	Mean	Variance	Mean	Variance	Mean	Variance
SoE	0.1411	0.0474	0.1317	0.0442	0.1411	0.0492	0.1726	0.0582
NAPS	0.5532	0.0713	0.5493	0.0764	0.5468	0.0743	0.5842	0.0794
HOS	0.2707	0.0609	0.2689	0.0649	0.2672	0.0644	0.2703	0.0594

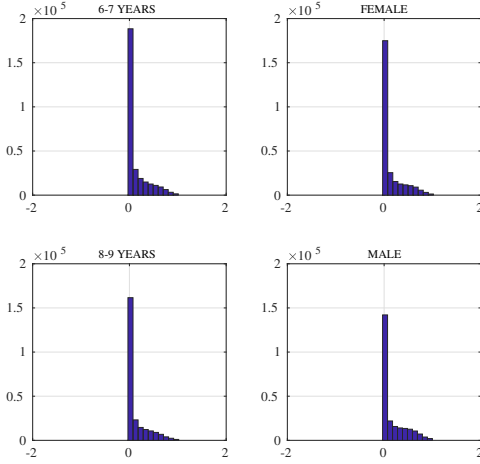


Figure 2: Histogram representing the variation of SoE parameter with respect to the age and gender of the child speakers.

ASR or not. For that purpose, we took speech data from 28 child speakers in the age-group 6-7 years, 45 child speakers in the age-group 8-9 years, 51 female children, and 24 male child speakers. This data was derived from the CMU kids speech database [16]. The glottal activity parameters were extracted from the considered speech data and histograms were derived for each of the parameters. The histograms for SoE are shown in Figure 2 while the mean and variance values for the same are given in Table 1. It is clearly visible from the histograms shown in Figure 2 that the SoE values do not vary much with a change in the age and gender of the speakers. Similar observations are noted in the case of NAPS and HOS. Since the performance of an ASR system should be independent of the age and gender of the speakers, similar mean values and lower variances suggest that the use of the explored glottal activity parameters will be suitable for the ASR task. We also performed canonical correlation analysis (CCA) between the MFCC features and GA parameters. CCA resulted in very low coefficients which suggests that the two kinds of features are highly uncorrelated and hence can be concatenated to capture a wider range of relevant acoustic information.

2.4. fMLLR-based normalization

Feature-space maximum likelihood linear regression is well known and one of the most frequently used normalization techniques [17]. It uses an affine transform for acoustic feature normalization. In the case of fMLLR, an alternate way to perform adaptation is to apply the transformation matrix on the observa-

tion sequence. The transformed observation vector, $\hat{\mathbf{O}}_t$ at any time t is given as:

$$\hat{\mathbf{O}}_t = \mathbf{A}_f^{-1} \mathbf{O}_t + \mathbf{A}_f^{-1} \mathbf{b}_f \quad (6)$$

where, $\mathbf{O}_t$ is the original observation vector at time t while $\mathbf{A}_f$ is $n \times n$ transformation matrix and $\mathbf{b}_f$ is an n -dimensional bias vector. In order to optimize the fMLLR transform, we performed speaker adaptive training (SAT) on a Gaussian-mixture-model-hidden-Markov-model (GMM-HMM) based ASR system. We have then concatenated the fMLLR-normalized MFCC features with GA parameters and fed the resulting feature vectors as the input for E2E system training. The use of fMLLR-based feature normalization helps in reducing the ill-effects of inter- and intra-speaker variations to a large extent.

3. Experimental evaluations

3.1. Database and experimental setup

For experimental evaluations, we have used two children's speech corpora, namely, CMU kids [16] and CSLU kids [18]. Children's speech training set was derived from the CMU and CSLU kids corpora which consisted for 55.02 hours of data from 1,193 child speakers. We resorted to in-domain data augmentation in order to increase the amount of training data. Consequently, a total of 493.36 hours of children's speech data was available for training purpose. Two different test sets were created for evaluation. A children's speech test set comprising of 2.66 hours of data was derived from the CMU kids corpus. This test set consisted of 1,539 utterances from 75 children with a total of 62,925 characters. Another test set was created from CSLU kids corpus. This test set consisted of 9.86 hours of data from 517 child speakers. There were a total of 10,000 utterances in this test set with 151,159 characters. To gain better insight into the effect of the proposed approach, the test set derived from CMU kids corpus was split into two based on the age of the speakers. The first split (**Age-6-7**) consisted data from children in the age group 6 to 7 years. The second split (**Age-8-9**) comprised of data from speakers belonging to the ages 8 to 9 years. Further to that, another splitting was done based on the gender of the speakers to create two more testing subsets.

Front-end speech parameterization was done using MATLAB. In the case of the MFCC features, speech data was analyzed into short-time frames using overlapping Hamming windows of duration 25 ms with a frame-shift of 10 ms. A 40-channel log-Mel-filterbank was used to compute the 13-dimensional MFCC feature vectors. The base features were time-spliced using a context size of ± 4 and then de-correlated, projected to 40 dimensional subspace and normalized using LDA, MLLT and fMLLR, respectively. For each of the short-time frames, the three glottal activity parameters were extracted.

The ESPnet [19] toolkit was used to develop Transformer-

Table 2: CERs and WERs for the children’s speech test sets demonstrating the effect of frame-level concatenation of MFCC features with glottal activity parameters.

Front-end features	CER (in %)		WER (in %)	
	CMU	CSLU	CMU	CSLU
MFCC (Baseline)	9.5	1.8	12.0	2.0
MFCC + HOS	7.3	1.7	9.5	1.9
MFCC + NAPS	7.5	1.8	9.5	2.0
MFCC + SoE	7.3	1.6	9.5	1.9

based E2E speech recognition systems. The Transformer-based E2E architecture had 12 encoder layers and 6 decoder layers. The dimension of attention was 256 and the number of attention heads was 4. The number of hidden units in each of the encoder and decoder layers was of 2048. Training was performed for a batch-size of 32 using the Noam optimizer. The label smoothing weight, gradient clipping, and accumulating gradients values were 0.1, 5 and 2, respectively. Training was performed for 20 epochs with a drop-out rate of 0.1. The language model (LM) comprised of single layer with 1000 units and it’s training was done using character-based recurrent neural network (RNN). The training of the LM was performed for 20 epochs. The batch size was chosen to be equal to 300. Stochastic gradient decent was employed for optimization. The out of vocabulary rate for the trained LM was equal to 0.0%. Performance evaluation metrics were character error rate (CER) and word error rate (WER).

3.2. Results and discussion

The CERs and WERs for the children’s speech test sets are given in Table 2. It is worth highlighting here that, CERs were extremely poor ($> 70\%$) when in-domain data augmentation was not performed since the available data was insufficient to train so many model parameters. In-domain data augmentation reduced the baseline CERs/WERs to level where those were similar to the performances reported in the recent state-of-the-art works [8, 9] including those employing large pre-trained models with fine-tuning [11, 12]. Furthermore, it is worth noting that the CERs as well as the WERs, in the case of CSLU kids test set, are much lower than those for the CMU kids test set. This is due to the fact that, the utterances in CSLU kids test set are short and majorly consist of single isolated words unlike the case of CMU kids. The E2E-based ASR systems are observed to perform poorly when the utterances are long [8]. At the same time, more than 90% of the training data is from the CSLU kids corpus. Hence, the CMU kids test set is a comparatively tougher test set. For both the test sets, WER and CER values are observed to decrease significantly when the glottal activity parameters are concatenated. However, larger reductions are noted in the case CMU kids test set. Next, the effect of fMLLR was studied. For that purpose, SoE parameter was concatenated with fMLLR-normalized MFCC features. The WERs and CERs for this study are given in Table 3. Finally, we performed a study to determine the age-group-specific and gender-specific recognition performances. The respective CERs and WERs are also given in Table 3. When fMLLR-normalized MFCC features and SoE parameter are concatenated, there are significant reductions in CER as well as WER for each of the cases. Similar trends are observed when fMLLR-normalized MFCC features are concatenated with either HOS or NAPS parameter. These results statistically substantiate the efficacy of the proposed ap-

Table 3: Age-group and gender-specific CERs and WERs for children’s speech test set derived from the CMU kids corpus.

Front-end features	Feature normalization	Speaker group	CER (in %)	WER (in %)
MFCC + SoE	None	Full test set	7.3	9.5
		Age-6-7	7.6	9.8
		Age-8-9	7.2	9.6
		Female	6.1	7.9
		Male	9.6	12.9
MFCC + SoE	fMLLR	Full test set	5.7	7.5
		Age-6-7	5.3	6.6
		Age-8-9	5.8	7.9
		Female	4.6	6.0
		Male	7.5	10.1

proach in the context of low-resource children’s ASR task.

Finally, we trained three different ASR systems. The first one (ASR-I) was trained using fMLLR-normalized MFCC features. The second one (ASR-II) was trained on un-normalized base MFCC features with SoE parameter appended to it. And the third system (ASR-III) was trained on fMLLR-normalized MFCC features with SoE parameter concatenated to it. Decoded outputs from these three systems were compared to determine if there were some underlying patterns. In other words, we tried to find out some of those sound units whose recognition improved as consequence of feature normalization and GA parameter concatenation. Though it is just an initial study and a more rigorous one is needed, yet we found some interesting patterns. For example, dental, labiodental, alveolar and bilabial sound units were better recognized by ASR-I compared to ASR-II. Similarly, recognition of nasal, alveolar and glottal sound units was better when ASR-II was used. As a consequence of this, when SoE parameters were concatenated with fMLLR-normalized MFCC features, error got further reduced. For example, the word OTTER was wrongly recognized majority of the times when ASR-II was used. However, it was correctly recognized 100 percent of the times by both ASR-I and ASR-III. Similarly, the recognition of the word MOUNTAINS failed majority of the time when ASR-I was employed. However, when ASR-II or ASR-III was used, the word MOUNTAINS got correctly recognized all the time. Similar trends were noted in the case of LIGHTNING, HOUSES and HOME.

4. Conclusion

The presented study aims at enhancing the performance of E2E speech recognition task for children in limited data conditions by concatenating fMLLR-normalized MFCC features with glottal activity parameters. To begin with, we have resorted to in-domain data augmentation techniques in order to achieve baseline recognition rates that are comparable to those obtained using the state-of-the-art techniques. Next, the effect of fMLLR-based feature normalization as well as of modeling glottal activity parameters was studied and documented. The explored glottal activity parameters are strength of excitation, normalized auto-correlation peak strength, and higher-order statistics. On concatenating these parameters with fMLLR-normalized MFCC features, statistically significant reductions in CER as well as WER are obtained.

5. References

- [1] S. Watanabe, T. Hori, S. Kim, J. R. Hershey, and T. Hayashi, “Hybrid ctc/attention architecture for end-to-end speech recognition,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 11, no. 8, pp. 1240–1253, 2017.
- [2] T. N. Sainath, Y. He, A. Narayanan, R. Botros, R. Pang, D. Rybach, C. Allauzen, E. Variani, J. Qin, Q.-N. Le-The, S.-Y. Chang, B. Li, A. Gulati, J. Yu, C.-C. Chiu, D. Caseiro, W. Li, Q. Liang, and P. Rondon, “An Efficient Streaming Non-Recurrent On-Device End-to-End Model with Improvements to Rare-Word Modeling,” in *Proc. INTERSPEECH*, 2021, pp. 1777–1781.
- [3] X. Chen, Y. Wu, Z. Wang, S. Liu, and J. Li, “Developing real-time streaming transformer transducer for speech recognition on large-scale dataset,” in *Proc. ICASSP*, 2021, pp. 5904–5908.
- [4] K. Kim, K. Lee, D. Gowda, J. Park, S. Kim, S. Jin, Y.-Y. Lee, J. Yeo, D. Kim, S. Jung *et al.*, “Attention based on-device streaming speech recognition with large speech corpus,” in *Proc. ASRU*, 2019, pp. 956–963.
- [5] L. Ye, G. Cheng, R. Yang, Z. Yang, S. Tian, P. Zhang, and Y. Yan, “Improving Recognition of Out-of-vocabulary Words in E2E Code-switching ASR by Fusing Speech Generation Methods,” in *Proc. INTERSPEECH*, 2022, pp. 3163–3167.
- [6] W. R. Huang, S.-Y. Chang, D. Rybach, T. Sainath, R. Prabhavalkar, C. Peyser, Z. Lu, and C. Allauzen, “E2E Segmenter: Joint Segmenting and Decoding for Long-Form ASR,” in *Proc. INTERSPEECH*, 2022, pp. 4995–4999.
- [7] L. Rumberg, H. Ehlert, U. Lüdtkke, and J. Ostermann, “Age-Invariant Training for End-to-End Child Speech Recognition Using Adversarial Multi-Task Learning,” in *Proc. INTERSPEECH*, 2021, pp. 3850–3854.
- [8] P. G. Shivakumar and S. Narayanan, “End-to-end neural systems for automatic children speech recognition: An empirical study,” *Computer Speech & Language*, vol. 72, p. 101289, 2022.
- [9] Ankita and S. Shahnawazuddin, “Developing children’s ASR system under low-resource conditions using end-to-end architecture,” *Digital Signal Processing*, vol. 146, p. 104385, 2024.
- [10] Ankita, Shambhavi, and S. Shahnawazuddin, “Effect of Modeling Glottal Activity Parameters on Zero-Shot Children’s ASR,” *IEEE/ACM Trans. Audio, Speech and Lang. Proc.*, p. 3039–3048, 2024.
- [11] R. Lu, M. Shahin, and B. Ahmed, “Improving children’s speech recognition by fine-tuning self-supervised adult speech representations,” *arXiv preprint arXiv:2211.07769*, 2022.
- [12] R. Jain, A. Barcovich, M. Yiwere, P. Corcoran, and H. Cucu, “Adaptation of whisper models to child speech recognition,” in *Proc. INTERSPEECH*, 2023.
- [13] T. Patel and O. Scharenborg, “Improving end-to-end models for children’s speech recognition,” *Applied Sciences*, vol. 14, no. 6, 2024.
- [14] K. S. R. Murty, B. Yegnanarayana, and M. A. Joseph, “Characterization of glottal activity from speech signals,” *IEEE signal processing letters*, vol. 16, no. 6, pp. 469–472, 2009.
- [15] E. Nemer, R. Goubran, and S. Mahmoud, “Robust voice activity detection using higher-order statistics in the lpc residual domain,” *IEEE Transactions on Speech and Audio Processing*, vol. 9, no. 3, pp. 217–231, 2001.
- [16] M. Eskenazi, J. Mostow, and D. Graff, “The CMU Kids Corpus LDC97S63,” <https://catalog.ldc.upenn.edu/LDC97S63>, 1997.
- [17] V. Digalakis, D. Rtischew, and L. Neumeyer, “Speaker adaptation using constrained estimation of gaussian mixtures,” *IEEE Transactions on Speech and Audio Processing*, pp. 357–366, 1995.
- [18] K. Shobaki, J.-P. Hosom, and R. Cole, “Cslu: Kids’ speech version 1.1,” *Linguistic Data Consortium*, 2007.
- [19] S. Watanabe, T. Hori, S. Karita, T. Hayashi, J. Nishitoba, Y. Unno, N. E. Y. Soplin, J. Heymann, M. Wiesner, N. Chen *et al.*, “Espnet: End-to-end speech processing toolkit,” *arXiv preprint arXiv:1804.00015*, 2018.